

Ahmed oueslati

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🎓 EDUCATION

Computer Science engineering
Insat

2020 – 2025

💼 PROFESSIONAL EXPERIENCES

Agentic AI and automation : House of Ichigo

2025/01
france

Marketing Intelligence & Data Pipeline Architecture:

- Developed, tracked and optimized marketing campaigns by centralizing data from multiple APIs (Brevo, Google Analytics, Clarity, Search Console) into a BigQuery warehouse with Cloud Storage
- Implemented Cloud Functions for data transformation with Cloud Scheduler for automated processing
- Created comprehensive dashboards with Looker Studio for real-time marketing performance monitoring

Multi-Agent System Development with n8n:

- Deployed a distributed multi-agent system in n8n using workflows for complete business process automation (business plan generation, go-to-market strategies)
- Built agents by scraping existing sources and creating structured vector store indexes
- Designed agent architecture including reasoning capabilities, context management, specialized tools, and optimized prompts

RAG Solutions & Vertex AI Integration:

- Designed an adaptive RAG agent using Google Agent Development Kit (Vertex AI) for intelligent querying of SEO documents stored in Google Drive
- Integrated specialized tools: document management, corpus creation, keyword search, and semantic analysis
- Deployed the solution in production via Cloud Build and Cloud Run with containerized Docker architecture

Data science Intern at TALAN

2024/06 – 2024/09

- Conducted in-depth literature reviews and evaluated state-of-the-art machine learning models, including U-Net, ResNet, and Vision Transformers, for biomarker detection accuracy and efficiency.
- Developed and rigorously tested computer vision models through comprehensive simulations and benchmarking, achieving an optimal biomarker detection accuracy of 92%.
- Designed a web-based interface integrating conversational AI to facilitate interactive data exploration and model interpretation for multidisciplinary stakeholders.
- Collaborated closely with cross-functional research teams to translate advanced generative AI techniques from research into practical, scalable solutions.

Ai Intern at SOFT PARDIGM

2023/07 – 2023/08

- Extracted text from resumes using Tesseract OCR and performed data cleaning and preprocessing.
- Annotated and labeled CV data using Label Studio for NLP model training.
- Trained and benchmarked spaCy Transformer models for semantic text mapping and information extraction.
- Developed a Flask-based API, deployed with Docker and integrated into Talentsoft software, improving extraction precision by 15%.

📁 PERSONAL PROJECTS

Implementation of a graph-transformer architecture for image classification: graph theory / Transformer

- Combined recent advances in graph processing and transformers, better capture spatial and structural relationships within wsl image data given a small dataset
- Designed a benchmarking protocol to evaluate state of the art CNN and ViT algorithms with our model.

Medical Visual Question Answering (MVQA) System (research)

- Compared image encoding techniques (Resnet, VGG..) and language embedding encoding (Lstm, BERT) and fusion techniques (BAN, concatenation, multi head attention)
- compared multiple benchmarking techniques (classification , BLEU for language génération) on predicting size, color, Modality , organ system , location, existence of abnormality
- Used multiple datasets like VQA-Med-2018 , VQA-Med-2019 , PathVQA

Serverless infrastructure in aws for computer vision

- Built an image processing pipeline leveraging AWS Lambda functions triggered by Amazon S3 events to perform real-time image analysis.
- Developed and deployed custom Lambda functions integrated with AWS SageMaker endpoints for ML-based image classification and scoring.
- Utilized AWS Step Functions to orchestrate workflow steps including image preprocessing, model inference, and result aggregation.
- Implemented comprehensive monitoring, logging, and alerting using AWS CloudWatch to ensure operational reliability and rapid incident response.

RAG Agent for Multi-Corpus Question Answering on GCP

- Designed and deployed a production-ready Retrieval-Augmented Generation (RAG) agent on Google Cloud Platform for semantic QA over custom corpora
- Leveraged Vertex AI PaLM 2 / Gemini models for rag_query tasks, integrated with LangChain and Chroma for vector database indexing
- Implemented modular agent tools: create_corpus, add_data, get_corpus_info, delete_corpus, with support for dynamic corpus updates via API

- Deployed ingestion pipeline using Cloud Functions and Pub/Sub, storing raw and processed files in Google Cloud Storage (GCS)
- Built a DevTool-like UI using FastAPI + Streamlit to trace agent decisions, request/response payloads, and tool usage in real time
- Ensured scalability and reproducibility with Docker, deployed on Cloud Run, and managed authentication & quotas via Service Accounts and IAM policies
- Focused on production-grade architecture, observability, and full integration with GCP-native MLOps stack

TECHNICAL SKILLS

Web Dev

- **Frontend** :HTML, CSS, JavaScript, Angular, Bootstrap
- **Backend** :Flask,NestJS,springboot
- **Databases**:MySQL, MongoDB, PostgreSQL
- **API** :Postman
- **Versioning tools** : Git, GitHub, GitLab

Cloud Computing

- **AWS** : Lambda, S3, EMR, DynamoDB, Step Functions
- **GCP** : BigQuery, Cloud Functions, IAM
- **Orchestration & Containerisation** : Docker, Kubernetes, Helm
- **Infrastructure as Code** : Terraform
- **Monitoring & Logging** : CloudWatch, Grafana, Prometheus
- **CI/CD** : Jenkins, GitLab CI

Data Engineering

- ETL : Apache Airflow, Spark
- Bases de données : PostgreSQL, MongoDB, MySQL, BigQuery
- Structuration & modélisation : Pandas, NumPy

CERTIFICATS

- Supervised machine learning:regression and classification 
- Advanced Learning Algorithms
- Efficient Large Language Model (LLM) Customization 

LANGUAGES

- English: professional
- French :Professional
- Arabe: Native
- Spanish : Beginner